

Day 15 Internship Report

PRODUCT COUNT ANALYSIS • SAMPLE SUPERSTORE DATASET

1. Executive Summary

The Day 15 task focused on generating product count analyses and establishing categorical hierarchies within the dataset. Using Excel pivot tables and specialized COUNTIFS/SUMIFS functions, the 9,994 transaction lines were categorized to highlight sales distribution, product line frequency, and total unique product offerings. This foundational audit establishes the baseline behavior of the product catalog.

2. Product Count Breakdown

CATEGORY	ORDER LINES	UNIQUE PRODUCTS	SALES REVENUE	SHARE OF LINES
Furniture	2,121	375	\$7,42,000	21.2%
Office Supplies	6,026	1,083	\$7,19,047	60.3%
Technology	1,847	404	\$8,36,154	18.5%

Category Definitions

The dataset is structurally divided into three main business operational lines: Furniture, Office Supplies, and Technology.

Order Lines (Volume)

Represents the number of individual times an item was purchased. Office Supplies dominates (6,026 lines), making it the most frequently bought category. Technology is the least frequent (1,847 lines).

Unique Products (Variety)

Counts the distinct individual products available within each category. Office Supplies offers the most variety (1,083 unique items), whereas Furniture has a much narrower catalog of just 375 distinct products.

Share of Lines

The percentage of total dataset transactions. It confirms that Office Supplies accounts for the vast majority (60.3%) of all everyday sales activity, compared to Furniture (21.2%) and Tech (18.5%).

i Key Business Insight: Revenue vs. Volume

A deeper look at the Sales metric reveals a classic retail split. Technology leads in total revenue at **\$8,36,154** despite having the absolute lowest transaction volume. Conversely, Office Supplies generated the lowest revenue (**\$7,19,047**) despite capturing over 60% of the volume. Office Supplies drives high-frequency, low-ticket sales (bringing in consistent orders), while Technology serves as the primary, low-frequency, high-ticket revenue engine.

3. Top Category Performance Hierarchy

Evaluating the "Top Category" requires strict metric definitions, as leadership diverges significantly depending on whether the business is optimizing for logistics/volume or financial revenue.

TOP CATEGORY (BY LINES) Office Supplies 6,026 Transactions	TOP CATEGORY (BY SALES) Technology \$8,36,154 Total	CATALOG SIZE (ALL) 1,862 Unique Products
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4. Multi-Criteria Audit (Edge Cases)

Further complex conditional counting was performed using multi-criteria array formulas to evaluate specific operational edge cases within the categories.

Technology lines sold at a loss	271 lines
Office Supplies lines distributed at a discount ≥ 40%	680 lines
Furniture lines in the West region generating profit > \$100	59 lines